YU XINTONG

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Education Background

09/2021-06/2025 Computer Science and Technology, School of Information Science and Engineering, Shandong Agricultural University (SDAU), China

Degree: Bachelor of Engineering

Arithmetic Score: 90.89 **Weighted Average Score:** 90.23

GPA: 4.02/5 **Ranking:** 27/195

Publication

Li Y, Zhang T, Yu X, Sun F, Liu P, Zhu K. Research on Agricultural Product Price Prediction Based on Improved PSO-GA. *Applied Sciences*. 2024; 14(16):6862. https://doi.org/10.3390/app14166862

Software Copyright and Patent

Computer software copyrights in China (copyright co-owner):

- Agricultural Intelligent Q&A System for Knowledge Graph Visualisation V1.0

Utility model patents in China:

- A waste shell rinsing and drying equipment

Research Experiences

Group Research:

12/2022-7/2024 Research on Optimal Aircraft Trajectory Based on the Combination of A* Search Algorithm and Monte Carlo Method

Tutor: Professor FAN Hao, SDAU

- Used the Monte Carlo Method to analyse aircraft trajectory planning
- Studied and improved the A* search algorithm and analysed its feasibility
- Simulated the trajectory planning path of the aircraft using MATLAB

12/2023-01/2024 Smart Agriculture - Automatic Soil Irrigation System

Tutor: Professor WANG Lu, SDAU

- Used sensors to collect environmental parameters of soil moisture
- Conducted data communication with CVM using the MQTT protocol, to turn on or off the sprinkler automatically

07/2023-11/2023 Intelligent Q&A System Based on Agricultural Knowledge Mapping

Tutor: Professor GUO Xuchao & Professor HAO Xia, SDAU

- Used Adobe XD and Sketch to design the page and develop the front-end interface
- Front-end, back-end, and database interaction
- Tested the system to verify its performance and accuracy

Achievement: First Prize of Shandong College Student Software Design Competition, 2023

Individual Projects:

02/2025-06/2025 Brain Tumor Identification and Prediction System Based on VGG19 and SVM

Tutor: Professor ZHANG Jijun, SDAU

- Applied convolutional neural network methods to MRI-based brain tumour identification and classification; used machine-learning methods for prediction.
- Developed a web-based identification and prediction system; awarded Outstanding Bachelor's Thesis at SDAU.

06/2023-07/2023 "Uniapp" Online Shopping Software Development

04/2023-05/2023 Course Assignment Management System Development

03/2023-04/2023 Business Card Management System Development

09/2022-10/2022 Library Management System Development

06/2022-06/2022 Contacts Management System Development

Honpurs and Awards

06/2025 Outstanding Undergraduate Graduate, SDAU

06/2025 Outstanding Bachelor's Thesis, SDAU

09/2024 Second Class Academic Scholarship, SDAU

04/2024 Third Prize, 15th Anniversary Lanqiao Cup National Software and Information Technology Professionals Competition

11/2023 First Prize, Shandong College Student Software Design Competition, Shandong Association for Science and Technology

11/2023 Third Class Academic Scholarship, SDAU

12/2022 Third Prize, 2023 University Physics Competition, Shandong Association for Science and Technology & Shandong Provincial Education Department

11/2022 Third Class Academic Scholarship, SDAU

Internships

09/2025-02/2026 Development Intern, Zhongying Qi'an (Shenzhen) IoT Technology Co., Ltd.

- Developed modular LLM-based multi-agent systems, integrating perception, memory, and decision-making modules.
- Implemented robot control logic using Python and ROS, supporting path planning, obstacle avoidance, and system integration.
- Tested and optimised AI agents and robotic systems to improve autonomous task execution and interaction capabilities.

12/2023-02/2024 Data Analysis Engineer, Shandong Juxin New Energy Technology Co., Ltd.

- Developed data processing workflows and SQL transformation scripts to organise, clean, and aggregate datasets for analysis.
- Applied SPSS and Vensim for data analysis, modelling, and evaluation of lithium-ion battery projects, producing reports to support project assessment.

07/2023-09/2023 Java Backend Developer, Shandong Liangzou City Development Group Co.

- Developed backend services and RESTful APIs using Java, supporting information management systems and automated business processes.
- Participated in database design, query optimisation, unit testing, and microservice-based system development to improve software maintainability.

01/2023-02/2023 Technology Development Intern, Shandong Zinuo Landscaping Co., Ltd.

- Assisted in developing smart irrigation and environmental monitoring applications, integrating IoT devices for data collection and system management.
- Applied Geographic Information Systems (GIS) for spatial data processing, mapping, and environmental analysis.

Extracurricular Activities

03/2024 A Study on the Management and Countermeasures of the Lower Yellow River Governance and Development

Details: We explored the current status of water quality based on collected data using Mann-Kendall (M-K), hierarchical clustering, and principal components analysis in conjunction with an improved composite Water Quality Index (WQI) and GIS.

08/2023 Smart Agriculture - Summer Rural Field Research, SDAU

Details: We collected and analysed large amounts of agricultural data using Python and Tableau and set up service stations to promote the development of smart agriculture by providing scientific information and technical support using big data and AI technologies.

Skills

- Computer language: Python, C++, Java, SQL
- Hardware and software management: jupyter notebook, Eclipse, PyCharm, VS Code
- Data Management: MySQL, Oracle, SQL Server
- AI Frameworks & Libraries: TensorFlow, PyTorch, OpenCV, Keras, Scikit-lear